

Smart Hostel Complaint Management System

IIT Roorkee Hostel Council Digital Infrastructure



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Problem Domain:

Hostel Administration & Student Services

Key Technologies:

HTML / CSS / JavaScript

Data Structures & Algorithms

Dashboard Analytics

Slide 2: Problem Understanding

Existing Challenges

Hostel residents frequently face issues related to:

- ✗ **Wi-Fi outages**
- ✗ **Electrical failures**
- ✗ **Plumbing problems**
- ✗ **Cleanliness concerns**
- ✗ **Infrastructure maintenance**

Problems with Existing System

- Complaints get lost or forgotten
- No prioritization mechanism (FIFO bottlenecks)
- Slow response times & lack of transparency
- No performance analytics or history tracking

Currently, complaints are managed through informal channels such as WhatsApp groups, phone calls, and manual registers.

OBJECTIVE

Develop a centralized digital complaint management platform that intelligently routes, prioritizes, tracks, and analyzes hostel complaints.

Smart Hostel Complaint Management System

01. Student Portal

Students can easily register complaints, track status in real-time, view their complaint history, and receive active updates from council members.

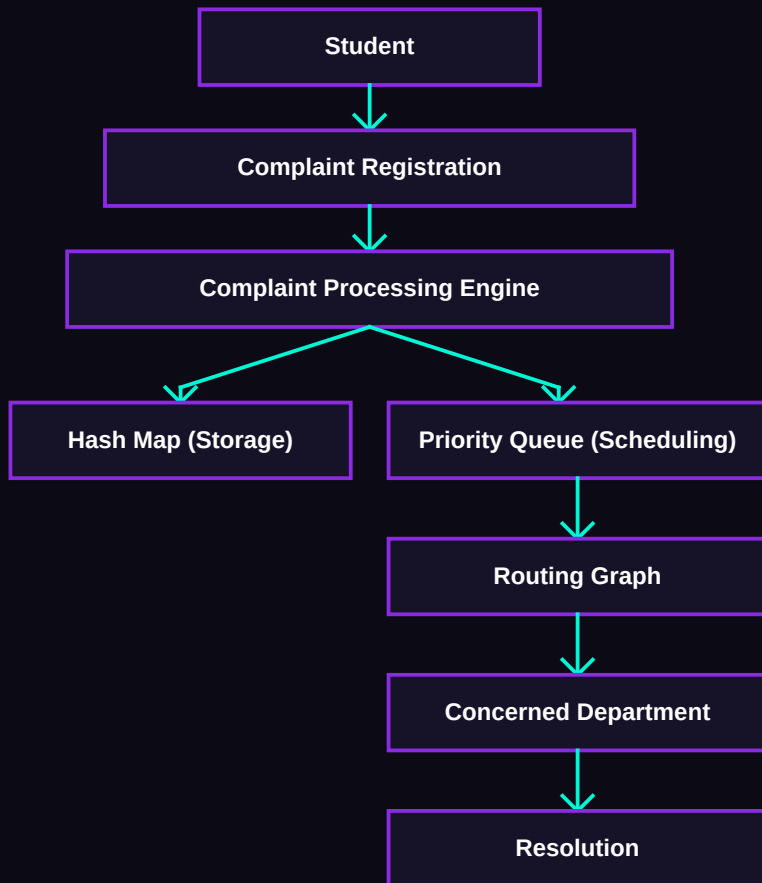
02. Council Dashboard

Council members can view all incoming complaints in priority order, dispatch tickets, update lifecycle states, and monitor workloads across hostels.

03. DSA Sandbox

An educational playground demonstrating how binary heaps, graphs, and hash maps function behind the scenes to optimize real-world operations.

Slide 4: System Architecture



Core Workflow Steps

1. Student submits complaint with description details.
2. Processing Engine performs simulated NLP classification.
3. Priority score (1-4) is assigned automatically.
4. Complaint is added to the Max-Heap Priority Queue.
5. Dijkstra's routing graph determines shortest path to dept.
6. Council processes complaint and chats with student.
7. Status updates and notifications are synced to user dashboard.

Slide 5: DSA Implementation Details

Data Structure	Implementation Details & Purpose	Complexities & Benefits
Priority Queue (Max Heap)	Prioritizes complaints based on severity. - Urgent Plumbing -> Priority 4 - Wi-Fi Outage -> Priority 3 - Fan Repair -> Priority 2 - General Feedback -> Priority 1	Complexities: - Insertion: $O(\log n)$ - Deletion: $O(\log n)$ - Peek: $O(1)$ Benefits: Fast complaint scheduling.
Hash Map (Chaining)	Instant complaint lookup mapping unique ticket IDs to objects. - COMP-2301 -> Details - COMP-2302 -> Details	Complexities: - Search: $O(1)$ - Insert: $O(1)$ Benefits: Fast retrieval, efficient status tracking.
Graph + Dijkstra Routing	Routes complaints to correct departments. - Path: Hostel -> Central Switch -> IT Department	Complexities: - Dijkstra: $O(E \log V)$ Benefits: Shortest path traversal, future scalability.

Student Portal

- Student registration form
- AI auto-categorization
- Complaint submission log
- Personal tracking dashboard

Council Dashboard

- Active priority queue dispatch
- Priority manual adjustments
- Complaint status management
- Hostel workload overview

Analytics Dashboard

- Total / pending / resolved counts
- Hostel caseload distribution
- Category workload graphs
- Resolution rate statistics

Expected Benefits Across Users

For Students

- > Faster complaint resolution times
- > Operational transparency on stages
- > Real-time chat threads & tracking

For Hostel Council

- > Better physical resource allocation
- > Reduced backlog via urgency queues
- > Data-driven dispatch decision making

For Administration

- > Real-time performance metrics
- > Proactive infrastructure monitoring
- > Historical complaint trend insights

Future Enhancements

- > **AI-Based Classification:** Automatically identify and route Wi-Fi, electrical, and other issues to the relevant department.
- > **Mobile App:** Native Android & iOS support for on-the-go notifications and problem reporting.
- > **Alert Notifications:** Auto-trigger email notifications, SMS alerts, and in-app notifications for updates.
- > **Predictive Maintenance:** Run analysis on historical reports to identify recurring issues and schedule preventive maintenance.

CONCLUSION

The Smart Hostel Complaint Management System transforms traditional hostel complaint handling into a data-driven, transparent, and efficient workflow.

By integrating Priority Queues, Hash Maps, and Graph Algorithms into real-world operations, the system not only improves complaint resolution but also demonstrates the practical application of Data Structures and Algorithms in solving campus-scale administrative challenges.